Lab Session

Task 1: Write a Python program to compute the dot product of vectors v ($\vec{v} = a\hat{\imath} + b\hat{\jmath} + c\hat{k}$) and u ($\vec{u} = d\hat{\imath} + e\hat{\jmath} + f\hat{k}$vectors in Python. a, b, c, d, e, f are constants and provided by the client.

Task 2: We will estimate the derivative of the function f(x) = -1/x at x=3 and check the result against the exact result.Tabulate your result. i.e, Error = Estimated value – True value. The results (output) are in the form:

Method	Numerical	Exact	Error
FDA	0.442829	0.447214	0.004385
CDA	0.447303	0.447214	0.000090
REA	0.447213	0.447214	0.000000
CDA2	-0.089487	-0.089443	0.000045
REA2	-0.089443	-0.089443	0.00000

Hint: Numerical derivatives are as follows:

FDA = (f(x + h) − f(x))/(h)

CDA = (f(x + h) − f(x − h))/(2h)

BDA = (f(xh) − f(x-h))/(h)

REA = (f(x − 2h) − 8f(x − h) + 8f(x + h) − f(x + 2h))/(12h)

CDA2 = (f(x − h) − 2f(x) + f(x + h))/h2

REA2 = (−f(x−2h)+16f(x−h)−30f(x)+16f(x+h)−f(x+2h))/(12h)

Task 3. Using Bisection Method and Newton-Raphson method, find the root of the function f(x) = x^3 -28 with accuracy at least 1%. Start with x0 = 2.

Algorithm (NR)

```
define f(x) = x^3 - 28
define f'(x) = 3 x^2
input x                          # input the initial root estimate
input Tolerance                  # input the tolerance (required accuracy)
do
Error = f(x) / f'(x)             # the error estimate
output x, Error                  # output current estimates
if ( |Error| < Tolerance )
terminate                        # terminate if tolerance is satisfied
x = x – Error                    # subtract the error estimate
```

TASK 4:Write a computer program to integrate the following functions to an accuracy of at least 6 decimal places using the Trapezoidal Formula and the Simpson's (1/3 and 3/8) Formula.

F(x) = (1-x^2)^(1/2) over the range x=-1 to x=1

F(t) = 894 t / (1.76 + 3.21 t^2)^3 over the range t=0 to t=1.61

Estimate the integrals from a rough sketch of the function.

HINT:

The Trapezoidal Rule is defined as : I = (b-a)/2) [f(a)+ (b)]

Simpson's 1/3 rule is defined as : I = ((b-a)/6) [f(a) + 4f(x1)+ f(b)]

Simpson's 3/8 rule is defined as : I = ((b-a)/8) [f(a) + 3f(x1)+ 3f(x2)+f(b)]